Week 4 — Visualisation, Power BI & Mini Project

Industrial Training TR-103 · Data Analytics Organisation: World With Web **Trainee:** Savreet Kaur · URN 2302668 **Period:** 6 July 2026 – 10 July 2026

Objective of the week

An analysis that nobody can read has not been delivered. This week covers presenting results: charting in Python, the principles behind choosing a chart, building an interactive dashboard in Power BI, and finally an end-to-end mini project combining everything from the four weeks.

Day 16 — Monday, 6 July 2026

Matplotlib

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(monthly["month"], monthly["sales"], marker="o")
ax.set_title("Monthly Sales, 2026")
ax.set_xlabel("Month")
ax.set_ylabel("Sales (₹)")
ax.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```

Chart types covered and what each is for:

CHART	SHOWS
Line	Change over time
Bar	Comparison across categories
Histogram	Distribution of one numeric column
Scatter	Relationship between two numeric columns
Box plot	Spread, median and outliers
Pie	Parts of a whole — used sparingly

Seaborn

Seaborn sits on top of Matplotlib with shorter syntax and better defaults for statistical charts:

```
import seaborn as sns

sns.boxplot(data=df, x="region", y="sales")
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Purples")
sns.scatterplot(data=df, x="discount", y="profit", hue="category")
```

The correlation heatmap from Week 2 was redrawn here, and the negative discount–profit relationship that was a number in a table became obvious at a glance — which is the argument for visualisation in one example.

Takeaway

Every chart needs a title, labelled axes and units. A chart without them forces the reader to guess, and they will guess wrong.

Day 17 — Tuesday, 7 July 2026

Choosing the right chart

The rule given was to start from the question, not from the chart library:

QUESTION	CHART
How has this changed over time?	Line
Which category is largest?	Horizontal bar, sorted
How is this spread out?	Histogram or box plot
Are these two related?	Scatter
What share does each part hold?	Stacked bar (pie only for 2–3 parts)

Common mistakes

- **Truncated y-axis on a bar chart.** Starting at a non-zero value exaggerates small differences. Bars encode length, so the axis must start at zero. Line charts are different — they encode position, so a truncated axis is acceptable there.
- **Pie charts with many slices.** Humans compare angles poorly. Beyond about three categories a sorted bar chart is easier to read.
- **Dual axes.** Two different scales on one chart can be arranged to imply almost any relationship.
- **3D effects.** They distort the very lengths and areas the chart exists to convey.
- **Rainbow colour scales.** They imply category boundaries that are not in the data. A single-hue scale is better for continuous values.

Dashboard design principles

1. One dashboard answers one question
2. Most important number top-left, where the eye lands first
3. Group related charts together
4. Consistent colours — one colour means one thing throughout
5. Leave white space; a crowded dashboard is not more informative
6. Label units and state the time period covered

Takeaway

Chart design is mostly about *not* misleading. The default settings of most tools are optimised to look impressive, not to be read accurately.

Day 18 — Wednesday, 8 July 2026

Power BI Desktop

The interface was introduced: the Report, Data and Model views, the Fields pane, the Visualizations pane, and Filters.

Getting data

Data was imported from three sources — Excel, CSV and a SQL database — to show that the connector changes but the rest of the workflow does not.

Power Query Editor

Transformations applied before the data reaches the model:

- Promote the first row to headers
- Set the correct data type on every column
- Remove unnecessary columns
- Replace values and handle nulls
- Split and merge columns
- Unpivot columns — turning a wide monthly layout into a tidy long one

Every step is recorded in **Applied Steps** on the right, so the sequence is visible, re-runnable on refreshed data, and removable if wrong. This is an advantage over cleaning in Excel by hand, where the steps leave no trace.

The data model

Separate tables were related rather than merged into one wide table:

- `Sales` (fact table) → `Products` (dimension) via `product_id`
- `Sales` (fact) → `Customers` (dimension) via `customer_id`
- `Sales` (fact) → `Calendar` (dimension) via `date`

This is the **star schema**, and it is the same relational thinking as the SQL joins of Week 3 — the difference is that the relationship is declared once in the model instead of written into every query.

Takeaway

Time spent in Power Query and the model view is repaid in every visual afterwards. Fixing the model later means rebuilding the visuals.

Day 19 — Thursday, 9 July 2026

DAX basics

DAX (Data Analysis Expressions) is the formula language of Power BI.

Calculated column — computed per row, stored in the table:

```
Line Total = Sales[quantity] * Sales[unit_price]
```

Measure — computed at query time, respecting the current filters:

```
Total Revenue = SUM(Sales[Line Total])

Total Orders = DISTINCTCOUNT(Sales[order_id])

Average Order Value = DIVIDE([Total Revenue], [Total Orders])
```

`DIVIDE` was recommended over the `/` operator because it handles division by zero without producing an error across the whole visual.

The distinction that mattered most: a **column** is fixed when the data refreshes; a **measure** recalculates for whatever slice the user has selected. Totals, averages and counts should almost always be measures.

Time intelligence

```
Revenue LY =
CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Calendar'[Date]))

YoY Growth % =
DIVIDE([Total Revenue] - [Revenue LY], [Revenue LY])
```

These require a proper date table marked as the date table in the model.

Building the dashboard

A sales dashboard was assembled:

- **KPI cards** — total revenue, total orders, average order value
- **Line chart** — revenue by month
- **Bar chart** — revenue by category, sorted descending
- **Map** — revenue by city
- **Table** — top 10 customers
- **Slicers** — date range, region, category

Interactions were configured so that selecting a bar filters every other visual, and the KPI cards update accordingly.

Takeaway

The slicers are what make a dashboard worth building. A static report answers the questions its author thought of; an interactive one lets the reader ask their own.

Day 20 — Friday, 10 July 2026

Mini project — end-to-end analysis

The final day combined all four weeks into a single piece of work.

Dataset: a retail sales dataset with roughly 10,000 order rows across customers, products, regions and dates.

Process:

STEP	TOOL	WEEK
1. Load and inspect	pandas	1
2. Clean — types, nulls, duplicates	pandas	2
3. Explore — distributions, correlation	pandas, Seaborn	2
4. Aggregate — revenue by region, category, month	SQL	3
5. Visualise — dashboard with slicers	Power BI	4
6. Present — findings and recommendations	—	4

Findings presented:

1. Revenue is concentrated — the top 20 per cent of customers account for a little over half of total revenue
2. Discounts above roughly 30 per cent turn profit negative, consistent with the correlation found in Week 2
3. One region underperforms on revenue despite a comparable order count, because its average order value is lower
4. Sales peak at year end, with the pattern repeating across the period

Recommendations: cap discounts at the point where profit turns negative, target the low average-order-value region with bundles rather than discounts, and plan stock for the year-end peak.

Training wrap-up

The project was presented to the trainer, followed by a review of the four weeks, questions, and feedback on the dashboard.

Takeaway

The analysis was the straightforward part. Deciding what mattered enough to show, and saying it in sentences a non-technical reader could act on, took longer than producing the numbers.

Summary of Week 4

DAY	DATE	TOPIC
16	Mon 6 Jul	Matplotlib and Seaborn
17	Tue 7 Jul	Choosing charts; dashboard design
18	Wed 8 Jul	Power BI, Power Query, data model
19	Thu 9 Jul	DAX measures, time intelligence, dashboard
20	Fri 10 Jul	Mini project and final presentation

Skills gained: charting with Matplotlib and Seaborn · choosing a chart that does not mislead · Power Query transformations · star-schema data modelling · DAX measures and time intelligence · building an interactive dashboard · presenting findings as recommendations.

Overall training outcome

WEEK	FOCUS	CORE SKILL
1	Python Basics	Load and inspect any dataset
2	Cleaning, Statistics, EDA	Fix the data and describe it honestly
3	SQL	Query a real database analytically
4	Visualisation & Power BI	Communicate the result

Across the four weeks the training covered the full path a data analyst follows: raw data in, cleaned and understood in the middle, a decision supported at the end. The mini project exercised every stage of it.